

Three Pāṇinian mechanisms

Input (English) with kāraka roles



(1) N-hot role embedding



$$W \cdot \mathbf{n} \oplus \mathbf{e}_{\text{tok}}$$

(2) Stratified masking



(3) śābdabodha head

role classifier
(per token)

discarded at inference

Transformer encoder · 14 L · 768 d · RoPE

$$\mathcal{L}_{\text{MLM}} + \lambda \mathcal{L}_{\text{Aux}}$$